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|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | A | <p>When the ammeter is zero,<br/> p.d. across the section of the wire on the left of the sliding contact is equal to that of the p.d. across the fixed resistor on the left.<br/> p.d. across the section of the wire on the right of the sliding contact is equal to that of the p.d. across the LDR on the right. (LDR is in bright light, its resistance is low)</p> <p>LDR in the dark has higher resistance, thus by potential divider principle, the p.d. across it will become higher.</p> <p>To return the ammeter reading to zero, we have to move the sliding contact to the left</p> |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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| 34 | D | The current in $D$ produces a magnetic field along the circumference of coil $C$ in the clockwise |
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